SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.2: Reflection and Congruence — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.2: Reflection and Congruence
Driving Question	DRIVING QUESTION: Why does a matatu's rear-view mirror show an image that is the same size and shape as the real object behind it — yet appears flipped — and how can we use this to prove that the original and its mirror image are always congruent?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Symmetry All Around Us — Anchoring the Matatu Mirror Phenomenon & Lines of Symmetry**	Students observe the anchoring phenomenon: a photograph of a matatu's rear-view mirror showing a flipped but same-sized image of a vehicle behind it. They also examine cut-out plane figures (circles, squares, Kenyan flag outline, leaf shapes) and fold them to locate where the two halves match perfectly.	A line of symmetry is a line that divides a plane figure into two congruent halves, each of which is a mirror image of the other. Simple figures may have one line of symmetry (e.g., an isosceles triangle), several (e.g., a square has four), or none (e.g., a scalene triangle). The matatu mirror acts as a physical line of symmetry between a real object and its reflected image.	Teacher note: Use the fold-test as the anchor for the entire sub-strand — if folding along a line produces a perfect overlap, that fold line IS the mirror line. Emphasise that 'same size and shape' is the intuitive entry point to congruence, which will be formalised in Lessons 7–8. Cold-call students to predict how many lines of symmetry the Kenyan flag has before they fold (expected answer: 0 — the shield design is asymmetric; this productive surprise motivates careful investigation). Pedagogical tip: Post the driving question on the classroom wall now and leave it unanswered; tell students every lesson will add one piece of the solution.
Lesson 2 **Reflection All Around Us — Locating Lines of Symmetry in Plane Figures**	Students systematically fold and draw lines of symmetry on printed outlines of an equilateral triangle, a rectangle, a regular hexagon, and a non-regular quadrilateral. They record their findings in a class table, noting the number of lines of symmetry for each figure.	An equilateral triangle has exactly 3 lines of symmetry; a rectangle has 2; a regular hexagon has 6; a non-regular quadrilateral typically has 0. A key generalisation: a regular polygon with n sides has exactly n lines of symmetry. Each line of symmetry is also a potential mirror line — the reflected half is an exact, flipped copy of the original half, previewing the properties of reflection	Teacher note: Guide students to notice the pattern between the number of sides of a regular polygon and its lines of symmetry before stating the rule — wait at least 10 seconds after posing the pattern question before cold-calling. Reinforce the link to the driving question by asking: 'If we placed a mirror along each symmetry line, what would we see?' Students should articulate

		explored in Lesson 4.	that the mirror image would complete the shape perfectly — same size, same shape, but flipped. This is the first verbal rehearsal of congruence language. Avoid introducing coordinate notation yet; keep this lesson purely geometric and visual.
Lesson 3 **Reflection on the Cartesian Plane — Plotting Object and Image Coordinates**	Students plot a simple triangle with labelled vertices on a Cartesian grid, then reflect it in (a) the y-axis, (b) the x-axis, and (c) the line y = x. They record object coordinates and image coordinates in a two-column table for each case.	Each mirror line produces a specific, predictable coordinate rule: reflection in the y-axis maps (a, b) → (−a, b); reflection in the x-axis maps (a, b) → (a, −b); reflection in the line y = x maps (a, b) → (b, a). These rules are derived from the data table, not memorised in isolation — students should be able to re-derive them from the pattern they observe. The Cartesian plane gives reflection a precise algebraic language that connects the visual flip seen in the matatu mirror to a mathematical rule.	Teacher note: Insist that students derive the coordinate rules from their own data tables before the teacher confirms them — this enacts the 'Observe → Explain' learning sequence. A common misconception is that reflection in y = x maps (a, b) → (−b, −a); address this explicitly by having students plot both options and check which one satisfies equidistance from the line y = x. Kenyan context tip: Frame the coordinate axes as 'roads' — the x-axis as Mombasa Road running east–west, the y-axis as Thika Road running north–south — to make quadrant navigation intuitive for Nairobi-based students.
Lesson 4 **Properties of Reflection — Discovering the Four Rules of the Mirror Line**	Using their plotted reflections from Lesson 3, students measure: (i) side lengths and angles of object vs. image triangles; (ii) perpendicular distances from each object point and its image to the mirror line; (iii) the angle at which the segment joining an object point to its image crosses the mirror line.	Reflection has four properties: (i) the image is the same size as the object (lengths and angles are preserved); (ii) object and image are equidistant from the mirror line (equal perpendicular distance); (iii) the segment joining any object point to its image is perpendicular to the mirror line; (iv) any point on the mirror line maps to itself (it is invariant). These four properties together define a reflection completely and will serve as the proof toolkit in Lesson 7.	Teacher note: Property (iv) — invariant points — is frequently overlooked by students; highlight it by asking 'What happens to a point that sits exactly on the mirror line?' (Wait 15 seconds; cold-call two students.) Emphasise that properties (i) through (iii) are what make reflected images congruent — size is preserved because lengths and angles are preserved. Connect explicitly to the driving question: 'We now have the mathematical reason why the matatu mirror image is the same size — can we now prove it is the same shape?' (Leave unanswered until Lesson 7.) Pedagogical tip: Have students write the four properties in their own words before seeing the formal statement.
Lesson 5 **Finding the Mirror Line — Determining the Equation of the Mirror Line Given an Object and Its Image**	Students are given pairs of congruent triangles on a grid (object and image) without being told the mirror line. They join corresponding vertices, find the midpoints of each joining segment, and draw the line through those midpoints. They then verify the line is perpendicular to each joining	The mirror line is always the perpendicular bisector of the segment joining any object point to its corresponding image point. To find its equation: (1) calculate the midpoint of the segment joining the object point to its image — this point lies on the mirror line; (2) calculate the gradient of that segment;	Teacher note: Students often confuse 'midpoint of the segment' with 'midpoint of the mirror line' — clarify that the mirror line extends infinitely; we are only finding one point on it. A useful pedagogical sequence: have students first verify numerically (by measuring distances on the grid) that their

	segment.	(3) use the negative reciprocal to find the gradient of the mirror line; (4) substitute the midpoint into y = mx + c to find the full equation. This reverses the process of Lesson 3 and consolidates the perpendicular bisector concept.	constructed line is equidistant from object and image before writing the algebraic equation — this anchors the algebra in the geometric property from Lesson 4. Kenyan context: Frame the task as 'You are a matatu driver; two vehicles are visible in your mirror. Where must the mirror glass be positioned?' — students find the mirror line as the answer.
Lesson 6 **Constructing Reflected Images — Drawing the Mirror Image of a Flag Outline on Plain Paper**	Students use a ruler and set square (no coordinate grid) to construct the reflection of a simplified Kenyan flag outline across an oblique mirror line drawn on plain paper. They apply the perpendicular bisector property point-by-point: drop a perpendicular from each key vertex to the mirror line, extend it an equal distance on the other side, and join the image points.	Every object point reflects to a unique image point located by the perpendicular bisector property: drop a perpendicular from the object point to the mirror line, then mark the image at the same perpendicular distance on the opposite side. This construction works for any mirror line orientation — horizontal, vertical, or oblique — and does not require a coordinate grid. The resulting image is congruent to the original object by construction, because every length and angle is preserved.	Teacher note: This is a practical construction lesson; accuracy matters. Insist students use a sharp pencil and check that their set-square corner is truly 90° before proceeding. A common error is measuring distance along the mirror line rather than perpendicular to it — demonstrate the difference explicitly with an exaggerated incorrect example first. Pedagogical tip: After construction, have students verify congruence by cutting out object and image and superimposing them (allowing a flip) — this tactile check directly answers the driving question at a concrete level and prepares students for the formal proof in Lesson 7.
Lesson 7 **Congruence Tests for Triangles — Proving the Matatu Mirror Image is Always an Exact Copy**	Students are given four pairs of triangles (each pair congruent by a different test: SSS, SAS, AAS, RHS) and one pair of non-congruent triangles. They measure sides and angles, then decide which congruence test — if any — applies to each pair. They then examine a reflected triangle from Lesson 3 and identify which test applies.	Two triangles are congruent if they satisfy any one of four tests: SSS (three sides equal), SAS (two sides and the included angle equal), AAS/ASA (two angles and a corresponding side equal), or RHS (right angle, hypotenuse, and one other side equal). A reflected triangle is always congruent to its object because reflection preserves all side lengths (SSS is automatically satisfied). This is the formal proof that the matatu rear-view mirror image is an exact copy of the real object — congruent, not merely similar.	Teacher note: Distinguish clearly between congruence (same size AND shape) and similarity (same shape, possibly different size) — students frequently conflate these. The key pedagogical move here is to apply SSS to the object–image pair from Lesson 3: all three sides are equal because reflection preserves length (established in Lesson 4, property i). This closes the logical chain started in Lesson 1. Cold-call students to state the full congruence proof in their own words (wait 20 seconds). Kenyan context: Use nyama choma cuts as an analogy — two pieces cut from the same template are congruent regardless of which way they face, just as a reflected triangle is congruent regardless of its orientation.
Lesson 8 **Final Explanation — Unifying Reflection and**	Students return to the original anchoring photograph of the matatu rear-view mirror from Lesson 1. They annotate it: labelling	Reflection maps every point according to a specific algebraic coordinate rule determined by the mirror line; the mirror line	Teacher note: This lesson is the synthesis lesson; resist the urge to re-teach content. Instead, use structured student

Congruence to Solve the Matatu Mirror Mystery	the mirror line (the glass surface), indicating the perpendicular bisector relationship, writing the coordinate rule that applies, and stating which congruence test proves the image is an exact copy of the real object.	is always the perpendicular bisector of the segment joining each object point to its image; reflection preserves all lengths and angles (the four properties); and the reflected image is therefore always congruent to the original object (provable by SSS). Together, these four ideas — coordinate rules, perpendicular bisector, preserved properties, and congruence tests — form a complete explanation of why the matatu rear-view mirror shows an image that is the same size and shape as the real object yet appears flipped.	presentations — each group annotates one aspect of the photograph (coordinate rule, mirror line equation, properties, congruence proof) and presents to the class. The teacher's role is to connect the presentations into one unified explanation. Pedagogical tip: End by returning to the driving question on the wall poster and cold-calling students to answer it in full sentences — insist they use the four technical terms: reflection, perpendicular bisector, preserved properties, and congruence. This lesson-8 full answer should be qualitatively richer than anything students could have said in Lesson 1, demonstrating measurable conceptual growth across the sub-strand.